TECHNICAL NOTES
LABOR FORCE SURVEY

II. Concepts and Definitions

a. Reference Period

The reference period for this survey is the “past week” referring to the past seven (7) days preceding the date of visit of the enumerator or the interviewer.

b. Employment Status Concepts

1. Population 15 Years Old and Over

This refers to number of population 15 years old and over excluding overseas workers. Overseas workers are excluded in the estimation of the size of working population (population aged 15 years and over) since the data on their economic characteristics are not collected because they are not considered part of the labor force in the country.

2. In the Labor Force or Economically Active Population

This refers to persons 15 years old and over who are either employed or unemployed in accordance with the definitions described below.

3. Employed

Employed persons include all those who, during the reference period are 15 years old and over as of their last birthday, and are reported either:

a. At work, i.e., those who do any work even for one hour during the reference period for pay or profit, or work without pay on the farm or business enterprise operated by a member of the same household related by blood, marriage, or adoption; or

b. With a job but not at work, i.e., those who have a job or business but are not at work because of temporary illness or injury, vacation, or other reasons. Likewise, persons who expect to report for work or to start operation of a farm or business enterprise within two weeks from the date of the enumerator’s visit are considered employed.

4. Underemployed

Underemployed persons include all employed persons who express the desire to have additional hours of work in their present job, or an additional job, or to have a new job with longer working hours. Visibly

underemployed persons are those who work for less than 40 hours during the reference period and want additional hours of work.

5. Unemployed

Starting April 2005, the new unemployment definition was adopted per NSCB Resolution Number 15 dated 20 October 2004. As indicated in the said resolution:

Unemployed persons include all those who, during the reference period, are 15 years old and over as of their last birthday, and reported as persons:

a) Without work, i.e., had no job or business during the reference period;

b) Currently available for work, i.e., were available and willing to take up work in paid employment or self-employment during the reference period, and/or would be available and willing to take up work in paid employment or self-employment within two weeks after the interview date; and

c) Seeking work, i.e., had taken specific steps to look for a job or establish a business during the reference period, or

Not seeking work due to the following reasons: (1) fatigued or believed no work available, i.e., discouraged workers; (2) awaiting results of previous job application; (3) temporary illness or disability; (4) bad weather; and/or (5) waiting for rehire or job recall.

6. Persons Not in the Labor Force

Persons 15 years old and over who are neither employed nor unemployed according to the definitions mentioned. Those not in the labor force are persons who are not looking for work because of reasons such as housekeeping, schooling, and permanent disability. Examples are housewives, students, persons with disability, or retired persons.

III. Sampling Design and Estimation Methodology

The LFS, being a household-based survey, uses the 2013 Master Sample (MS) design. The regular LFS (January, April, July and October LFS rounds) has four replicates equivalent to a total of 42,768 secondary sampling units (SSUs) or sample housing units while the special LFS (February, March, May, June, August, September. November and December LFS rounds) has one replicates equivalent to a total of 10,692 secondary sampling units (SSUs) or sample housing units are included as samples. Using a two-stage cluster sampling design, Enumeration Areas (EAs)/barangays are selected at the initial sampling stage as the primary sampling

units (PSUs), while the housing units within the selected PSUs are selected as the SSUs. Generally, all households within the sample housing unit are also considered as sample households. However, for housing unit with more than three (3) households, a maximum of three (3) sample households are randomly selected.

Sampling Frame

The 2013 MS sampling frame was constructed based on the results of the 2010 Census of Population and Housing. This was refreshed with the 2015 Census of Population results where the EA Reference File (EARF) was used as the PSU frame and the 2015 list of households for each of the PSUs were used as the SSU frame.

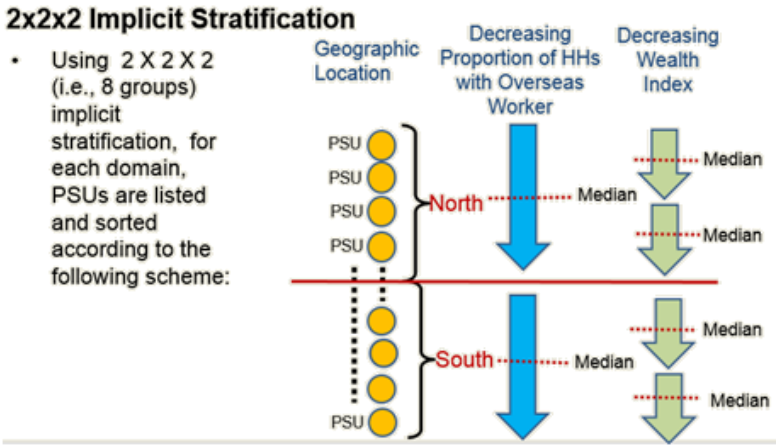
Sampling Domain

To provide subnational or provincial level statistics with precise estimates, the 2013 MS has 117 major domains as follows: 81 provinces (including the newly created province Davao Occidental); 33 highly urbanized cities (HUCs) (including 16 cities in the National Capital Region); and 3 other areas (Pateros, Isabela City, and Cotabato City).

Primary Sampling Units

In the 2013 Master Sample Design, each sampling domain (i.e., province/HUC) is divided into exhaustive and non-overlapping area segments known as PSUs with about 100 to 400 households. Thus, a PSU can be a barangay/EA or a portion of a large barangay, or two or more adjacent small barangays/EAs.

2x2x2 Implicit Stratification



The PSUs are then ordered according to the following: (1) North-South/West-East Geographic location; (2) Decreasing Proportion of HHs with Overseas Worker; and (3) Decreasing Wealth Index.

Replicates

The 117 sampling domains were represented in each replicate. A replicate is composed of ordered list of PSUs. Most of the provinces, that is, 75 out of 81, have six PSUs per replicate while in HUCs, eight PSUs form a replicate. Small domains, namely Batanes, Guimaras, Siquijor, Camiguin, Apayao, and Dinagat Islands have three PSUs per replicate.

Sample Allocation Scheme

The total number of sample SSUs is allotted proportionately to the measure of size of the PSU. Thus, a PSU with only 100 HHs has less number of sample HHs than PSUs with 400 HHs but, on the average, there are 12 sample HHs allotted for each PSU in HUCs and an average of 16 sample HHs for every PSU in the province.

A total national sample of 10,692 sample HHs is allotted for the monthly rounds of the LFS.

Domain	**1 Sample Replicate (National Level Estimate)**		**4 Sample Replicates (National and Regional Level Estimate)**	
	Number of Sample PSUs	**Number of Sample Housing Units/HHs**	**Number of Sample PSUs**	**Number of Sample Housing Units/HHs**
75 Province Domain (16 SSUs per PSU)	6	96	24	384
6 small provinces (Batanes, Guimaras, Siquijor, Camiguin, Apayao, and Dinagat Islands) (16 SSUs per PSU)	3	48	12	192
31 HUCs (12 SSUs per PSU)	8	96	32	384
2 small HUCs (12 SSUs per PSU)				
San Juan City	3	36	12	144
Lucena City	5	60	20	240
3 other urban areas (12 SSUs per PSU)				
Pateros	3	36	12	144
City of Isabela	3	36	12	144
Cotabato City	5	60	20	240
Total	**735**	**10,692**	**2,940**	**42,768**

Base weight computation

The base weight is computed as the inverse of selection probability.

$$w_{p\alpha\beta} = \frac{A_p}{\alpha_p} \; x \; \frac{B_{p\alpha}}{b_{p\alpha}}$$

where:

A_p - total number of PSUs in domain p

a_p - total number of sample PSUs in domain p

$B_{p\alpha}$ - total number of housing units in PSU α in domain p

$b_{p\alpha}$ - total number of sample housing units in PSU α in domain p

For housing units with at most 3 households the base weight is computed as

$$w_{p\alpha\beta} = \frac{A_p}{\alpha_p} \; x \; \frac{B_{p\alpha}}{b_{p\alpha}}$$

For housing units with more than 3 households the base weight is computed as

$$w_{p\alpha\beta y} = \frac{A_p}{\alpha_p} \; x \; \frac{B_{p\alpha}}{b_{p\alpha}} \; x \; \frac{C_{p\alpha\beta}}{c_{p\alpha\beta}}$$

where:

$C_{p\alpha y\beta}$ - total number of households in the sample housing unit

$c_{p\alpha y\beta}$ - the number of sample households in the sample housing unit=3

Base Weight Adjustment

The base weight is adjusted for unit non-response and further calibrated to conform to the known or projected population count. The projected population count used for May 2023 LFS was the May 2023 projected population.

For unit non-response adjustment (within domain p), the adjustment is computed as:

$$A_{p1} = \frac{\text{weighted}^* \text{ total number of eligible sample households}}{\text{weighted}^* \text{ total number of responding households}}$$

Where weighted * refers to the base weight.
Applying this to the base weight, we have:

$$w'_{p\alpha\beta}{}^{adj} = w_{p\alpha\beta} \; x \; A_{p1}$$

Further calibration is made to conform with known population count by age-sex as follows:

Age Group (in years)	Sex	
	Male	Female
0 – 14	C1	C2
15 – 24	C3	C4
25 – 34	C5	C6
35 – 44	C7	C8
45 – 54	C9	C10
55 – 64	C11	C12
65 and over	C13	C14

$$A_{p2c} = \frac{X_{pc}}{\hat{X}_{pc,adj}}$$

where:

X_{pc} - is the projected total population for age-sex class c

$\hat{X}_{pc,adj}$ - is the weighted estimate of the population for age-sex class c using the non-response adjusted weight

Hence the final weight (calibrated weight is):

$$w'_{p\alpha,fin} = \underbrace{w'_{p\alpha,adj}}_{\text{non-response adjusted weight}} \; x \; \underbrace{A_{p2c}}_{\text{population adjustment factor}}$$

Estimation of Totals

- For domain total

  The estimate for the population total for a domain (province/HUC) is derived using:

$$\hat{Y}_p = \sum_{\alpha=1}^{a_p} \sum_{\beta=1}^{b\alpha} w'_{p\alpha,fin}\, y_{p\alpha\beta}$$

- For the regional total (if domain is below regional)

  The estimate for the population total for the region is derived as the sum of the estimated totals of its provinces/HUCs which is given as:

$$\hat{Y}_r = \sum_{p=1}^{m_r} \left(\hat{Y}_p \right) = \underbrace{\hat{Y}_1 + \hat{Y}_2 + ... + \hat{Y}_{m_r}}_{\text{Weighted Province/HUC Totals}}$$

  where

  $\hat{Y}_p$ - estimated total for province/HUC *p*

  m_r - total number of provinces/HUCs in the region

- For the national total

  The estimate for the population total at the national level is derived as the sum of the estimated regional totals which is given as:

$$\hat{Y} = \sum_{r=1}^{n} \left(\hat{Y}_r \right) = \underbrace{\hat{Y}_1 + \hat{Y}_2 + ... + \hat{Y}_n}_{\text{Weighted Region Totals}}$$

  where

  $\hat{Y}_r$ - estimated total for region *r*

  n - total number of regions in the country

Estimation of Proportions/Ratios

The estimation of a population proportion or ratio of the formula R = Y/X where Y and X are population totals for variables y and x, respectively, is derived using the formula $\hat{R} = \frac{\hat{Y}}{\hat{X}}$.

Estimation of Sampling Error

Sampling error is usually measured in terms of the standard error for a particular statistic (total, mean, percentage, etc.), which is the square root of the variance.

If the samples are selected using simple random sampling, it would have been possible to use straightforward formulas for calculating sampling errors. However, the LFS is the result of a multi-stage design, hence it is necessary to use more complex formulas.

Sampling errors are computed using statistical programs. These statistical programs use the Taylor linearization method to estimate variances for survey estimates of means, proportions, or ratios.

a. Sampling Error for Totals

$$SE(\hat{Y}_p) = \sqrt{\hat{V}(\hat{Y}_p)}$$

$$\hat{V}(\hat{Y}_p) = (1-\frac{a_p}{A_p})a_p s^2 + \frac{a_p}{A_p}\sum_{\alpha=1}^{a_p}(1-\frac{b_{p\alpha}}{B_{p\alpha}})b_{p\alpha}{s_\alpha}^2$$

where:

$s^2 = \frac{\sum_{\alpha=1}^{a_p}(y_{p\alpha}-\frac{\widehat{y_p}}{a_p})^2}{a_p-1}$ $\qquad$ ${s_\alpha}^2 = \frac{\sum_{\beta=1}^{b_{p\alpha}}(y_{p\alpha\beta}-\frac{y_{p\alpha}}{b_\alpha})^2}{b_\alpha-1}$

$y_{p\alpha} = \sum_{\beta=1}^{b_{p\alpha}} {w'}_{p\alpha,fin}\, y_{p\alpha\beta}$ is the weighted total for psu α.

b. Sampling Error for Proportions or Ratios

The Taylor series linearization method is used to estimate the variance of a proportion or a ratio.

Its formula is given as follows:

$$SE(\hat{R}_p) \approx \frac{1}{{\hat{X}_p}^2}\left[\hat{V}(\hat{Y}_p) + {\hat{R}_p}^2\hat{V}(\hat{X}_p) - 2\hat{R}_p c(\hat{Y}_p, \hat{X}_p)\right]$$

Where:

$$\hat{R} = \frac{\hat{Y}}{\hat{X}}$$

$$c(\hat{Y}, \hat{X}) = \frac{a_p}{a_{p-1}}\sum_{\alpha=1}^{a_P}\left(y_\alpha - \frac{\hat{Y}}{a_p}\right)\left(x_\alpha - \frac{\hat{X}}{a_p}\right)$$

$$a_p = number\ of\ sample\ PSUs\ in\ domain\ p$$

In the LFS, the 117 province/HUC domains are also treated as natural stratification while the PSUs are treated as clusters.

Data Checking, Coding and Filtering Prior to Estimation of Proportions

Enumeration is a highly complex operation, and it may happen that reported/encoded entries during data collection have some omissions, and implausible/inconsistent entries. Editing is a process meant to correct these errors.

During the interview, embedded editing was activated, and errors/inconsistent entries were detected by the program. Editing was also done using Computer Aided Field Editing (CAFE) program after every interviewed household to ensure completeness and consistency of encoded entries. For monitoring of the status of data collection,

LFS raw data from the tablet is uploaded to the PSA Central Office server as soon as the interview of a household/EA was completed.

Review and verification of the PSOC and PSIC codes and invalid values for LFS data items were done in the provincial office using the LFS Information System (LFS IS). Further processing in the regional office such as ID validation, and completeness check, edit and matching of LFS sample households with the original List from Master Sample (MS) Form 6 were done to ensure that the number of households listed was fully covered.

Preliminary and final tabulations of data were done at the PSA Central Office.

IV. Dissemination of Results

The LFS preliminary results press release, and the statistical tables are publicly available at the PSA website www.psa.gov.ph. The final estimates of the will be released through the Statistical tables, six months after the data collection.